

17. Implement Mobile Price Mobile Price Prediction using Machine Learning

Aim

To implement a **Mobile Price Prediction Model** using an appropriate machine learning algorithm in Python and predict the price range of a mobile phone based on its features.

Algorithm

1. Import the required libraries.
2. Create a dataset containing mobile phone features and price ranges.
3. Separate the input features and target price range.
4. Split the dataset into training and testing data.
5. Create a **Decision Tree Classifier**.
6. Train the model using the training data.
7. Predict the price range of the test data.
8. Calculate the accuracy of the model.
9. Predict the price range of a new mobile phone.

Code:

```
from sklearn.tree import DecisionTreeClassifier

from sklearn.model_selection import train_test_split

from sklearn.metrics import accuracy_score
```

```
X = [
    [2, 32, 3000, 3000],
    [3, 64, 4000, 4000],
    [4, 64, 4500, 5000],
    [4, 128, 5000, 6000],
    [6, 128, 6000, 8000],
    [6, 256, 7000, 10000],
    [8, 256, 8000, 12000],
```

```
[8, 512, 10000, 15000],  
[12, 512, 12000, 20000],  
[16, 512, 15000, 25000]  
]
```

```
y = [  
    "Low",  
    "Low",  
    "Medium",  
    "Medium",  
    "Medium",  
    "High",  
    "High",  
    "High",  
    "Premium",  
    "Premium"  
]
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)
```

```
model = DecisionTreeClassifier(random_state=42)
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Actual Price Range:", y_test)
```

```
print("Predicted Price Range:", y_pred)
```

```
print("Accuracy:", accuracy * 100, "%")
```

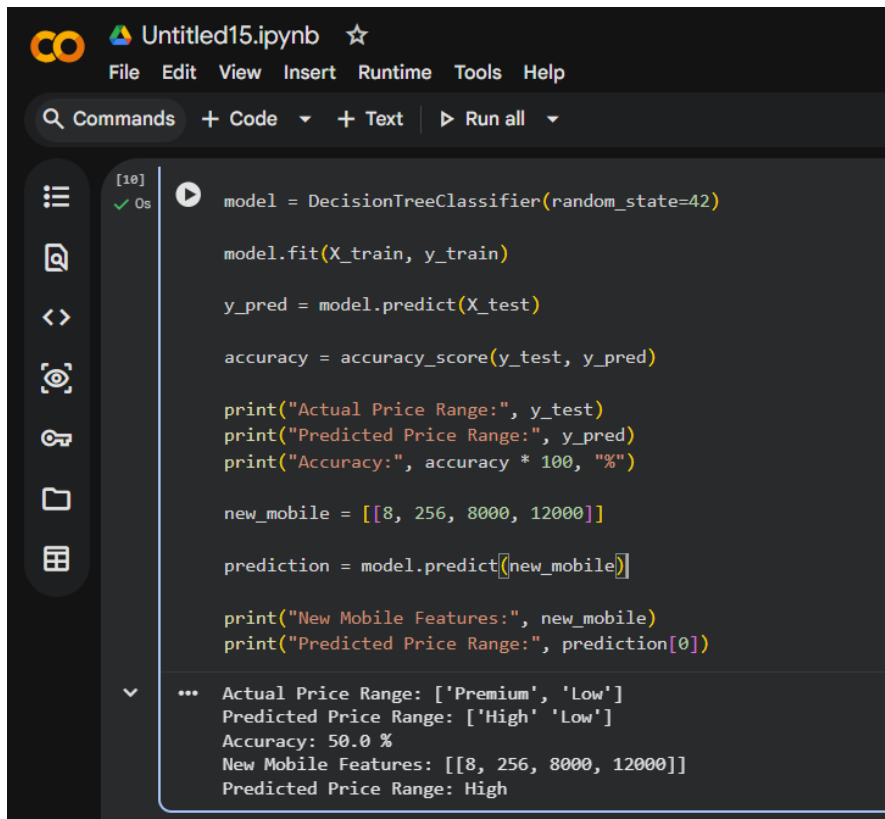
```
new_mobile = [[8, 256, 8000, 12000]]
```

```
prediction = model.predict(new_mobile)
```

```
print("New Mobile Features:", new_mobile)
```

```
print("Predicted Price Range:", prediction[0])
```

Output:



The screenshot shows a Jupyter Notebook titled 'Untitled15.ipynb'. The interface includes a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. Below the menu is a toolbar with 'Commands', '+ Code', '+ Text', and 'Run all'. The left sidebar contains icons for file management and viewing. The main area displays a code cell with the following Python code:

```
[10]
✓ 0s
model = DecisionTreeClassifier(random_state=42)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Actual Price Range:", y_test)
print("Predicted Price Range:", y_pred)
print("Accuracy:", accuracy * 100, "%")

new_mobile = [[8, 256, 8000, 12000]]

prediction = model.predict(new_mobile)

print("New Mobile Features:", new_mobile)
print("Predicted Price Range:", prediction[0])
```

Below the code, the output is displayed:

```
... Actual Price Range: ['Premium', 'Low']
Predicted Price Range: ['High' 'Low']
Accuracy: 50.0 %
New Mobile Features: [[8, 256, 8000, 12000]]
Predicted Price Range: High
```

Result Statement

Thus, the **Mobile Price Prediction** model was successfully implemented using the **Decision Tree Classifier**, and the price range of a new mobile phone was successfully predicted.